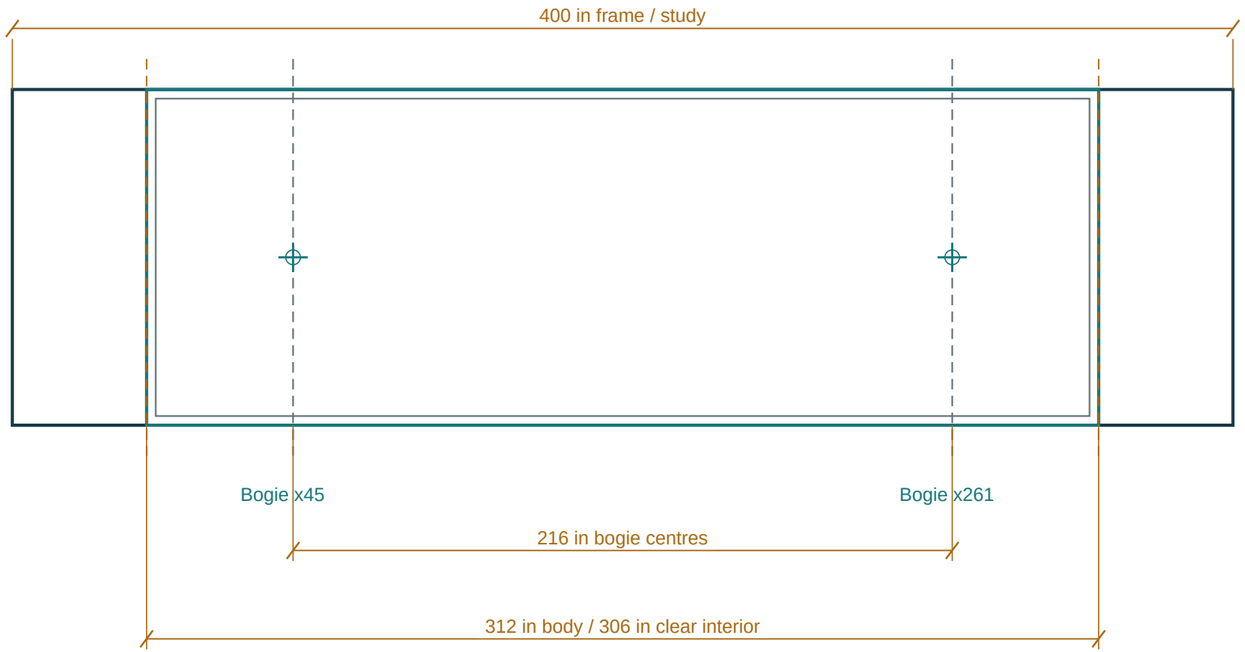
First-coach integration: interfaces and release gates

Phase 3C develops physical construction studies, supplier interfaces and first-article checks. All dimensions marked STUDY remain subject to professional review.

G1 / COACH CONTROL PLAN - RETAINED ENVELOPE



x0 = front inner wall; y0 = centreline; rail z0; finished floor z28 STUDY.

G2 / WHAT THIS ISSUE ADDS

A member grid and floor support trial; actual floor finish / bedding stack; body-panel joints and drainage principles; boarding interfaces; brake control architecture; jig sequence and instrumented verification.

No supplier-rated bogie, attachment design or approved passenger assembly is claimed.

G3 / COORDINATED HOLD POINTS

The frame mass and vertical stack must be reconciled before procurement. The 12 in girder is a stiffness study; side sills, crossmembers and nodes require full structural analysis.

No 3C drawing supplies missing pin diameters, weld sizes, axle profiles, joint preload or production tolerances.

G4 / RETAINED REQUIREMENTS

One locomotive; three regular coaches plus one accessible coach. 36 in gauge remains evaluation basis. 0% longitudinal grade is the owner target. Both stations need accessible boarding and evacuation planning; North Pole location remains deferred.

G5 / COMMON LOAD INPUT

Upper analysis allocation: regular13,200 kg each; accessible11,000 kg; locomotive + hotel14,500 kg; total65,100 kg. These are coordinated calculation cases, not actual weighed masses, approved ratings or permission to increase tare.

G6 / FIRST PHYSICAL WORK

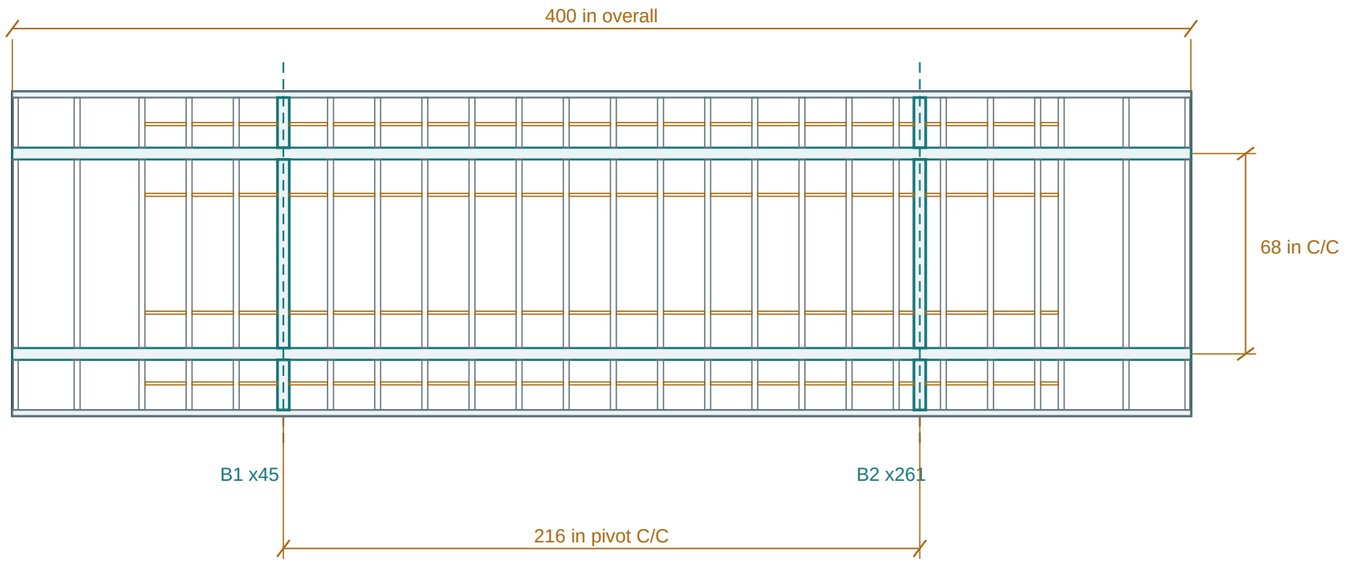
Start with stable floor-level seating and doorway mockups. A steel first article, rolling tests and pressure tests need released specialist procedures. Farm crew can mark geometry and keep records; production welds and loaded testing need assigned competent specialists.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Frame grid, bolster nodes and packaging section

STUDY: main12 x4 x1/4 HSS; floor crossmembers / side sills4 x2 x3/16 HSS. Member designation does not release material grade, cutting or welding.

F1 / FULL FRAME PLAN - CANDIDATE MEMBER WIDTHS

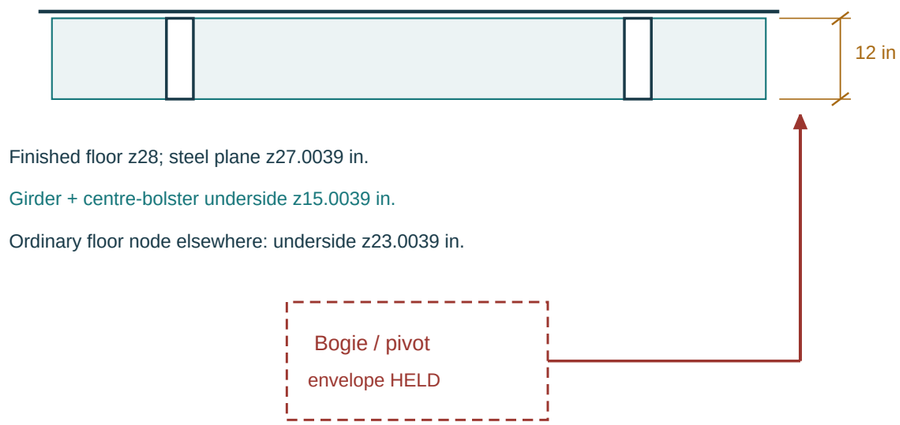


Main girders y=-34,+34. Side sills y=-54,+54. All top faces at steel plane.

Gold:3 x1 x1/8 seat-attachment rail candidates aty=-44,-20,+20,+44.

166 member envelopes total; all joints, anchors and pivot seats HELD.

F2 / TRANSVERSE SECTION AT BOLSTER



Finished floor z28; steel plane z27.0039 in.

Girder + centre-bolster underside z15.0039 in.

Ordinary floor node elsewhere: underside z23.0039 in.

Support top + interface + movement clearance <=15.0039 in.

All terms HELD; a shallower girder alone does not raise bolster.

F3 / CROSSMEMBER CENTRE STATIONS / INCHES

Cabin: -3, 13, 29, 45(B1), 61, 77, 93, 109, 125, 141, 157, 173, 189, 205, 221, 237, 253, 261(B2), 269, 285, 301, 309. Platforms: -46, -25, 331, 352.

Each ordinary transverse station has three clear segments: y-53..-36, y-32..32, y36..53. Nominal x width2. Bolster station width4. These are model envelopes; fit-up, coping, node plates and weld gaps are not defined.

F4 / NODE / LOAD-PATH HOLD

Bolster and drawbar forces must spread into the main girders through engineered joints. Butt-ended member envelopes do not establish load transfer.

16 in floor spacing is a trial grid, not a panel span approval. Check seat / wall anchors independently.

Beam-only mass 1,913.26 kg: 413.26 kg above the 1,500 kg allowance before held joints, draft structure and stairs.

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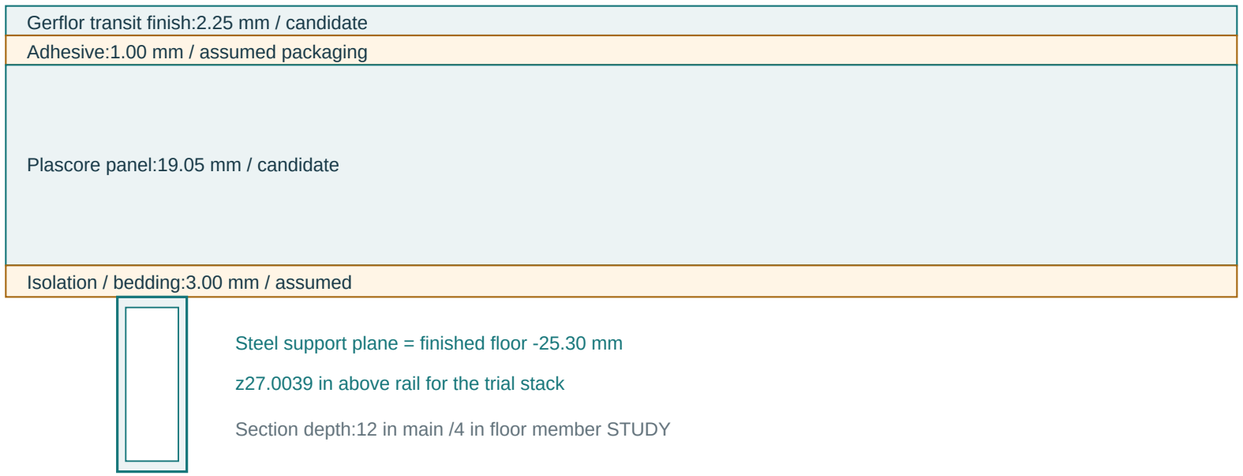
Floor, wall and roof: coordinated construction studies

Detail layers show verified product thicknesses and explicit packaging allowances. Load ratings, bonding schedule, fire performance and structural attachment remain HELD.

D1 / FLOOR SECTION - VERTICAL EXPANSION FOR CLARITY

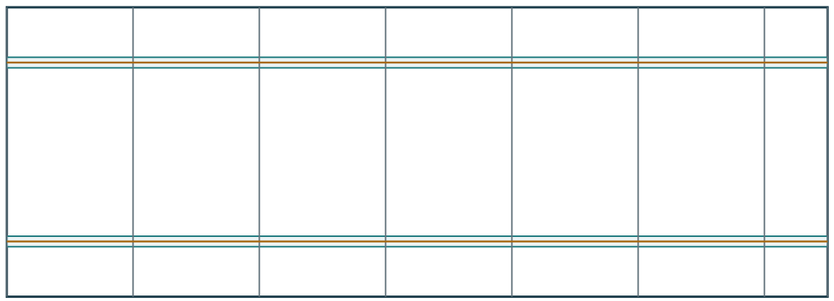


Steel support plane = finished floor -25.30 mm

z27.0039 in above rail for the trial stack

Section depth:12 in main /4 in floor member STUDY

D2 / FLOOR PANEL SEAMS - SUPPORTED GRID STUDY



21 panels from4 x8 stock

21 /68 /21 in strips

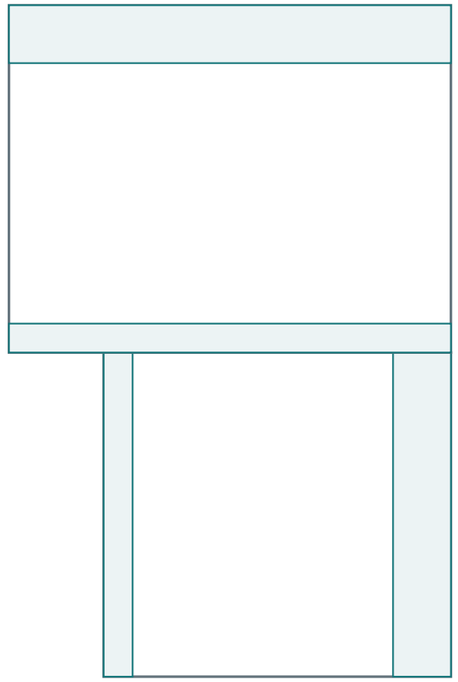
Seams aligny+/-34 girders

Alternate:7 fullwidthpanels

from5 x12 source stock

Panel seam support is a study. Seal cut honeycomb edges. Engineer bearing, joints and hard points for seat / wall / equipment anchors. Verify point loads and tested bonding system before production.

D3 / SIDEWALL + ROOF CORNER / ENVELOPE



Roof crown / curb

and cap HELD

Outer.50 in

Cavity2.25 in

Inner.25 in

Wall inner y52

Wall outer y55

Diagram is enlarged construction zoning, not a joint fabrication detail.

D4 / HEIGHT / WEATHER PATH

Clear ceiling z114.6142 in; planar roof outer z117.6142 in, before crown, trim, curb or HVAC. Seal outer laps with supplier-compatible products; drain outward. Do not drain into the insulation cavity or onto brake equipment.

Cavity must contain selected structure and insulation; 2.25 in is not an approved insulation thickness or wall framing specification.

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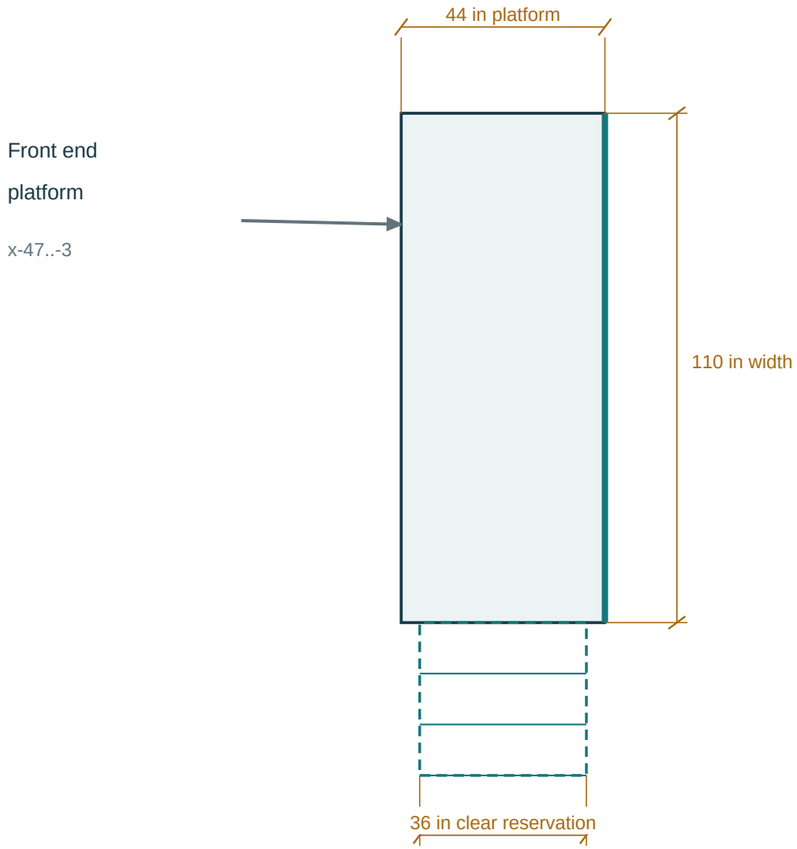
Boarding: platform stair and accessible bridge interfaces

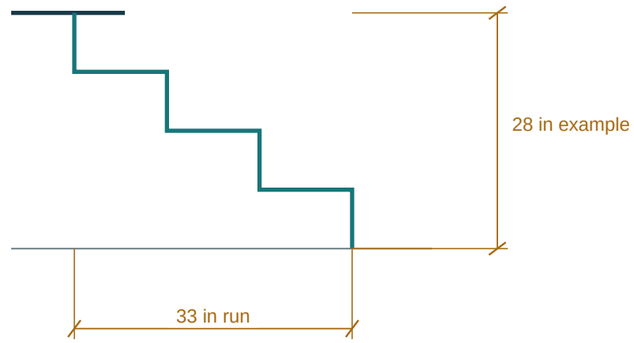
Candidate geometry is reserved for fit studies only. Applicable vehicle accessibility, emergency egress and station requirements must be established before final dimensions are selected.

A1 / END PLATFORM + DEPLOYED SIDE STAIR / PLAN



Only8 in total remains alongx for edges / posts / frame: fit HELD.

A2 / STAIR SIDE ELEVATION / EXAMPLE ONLY

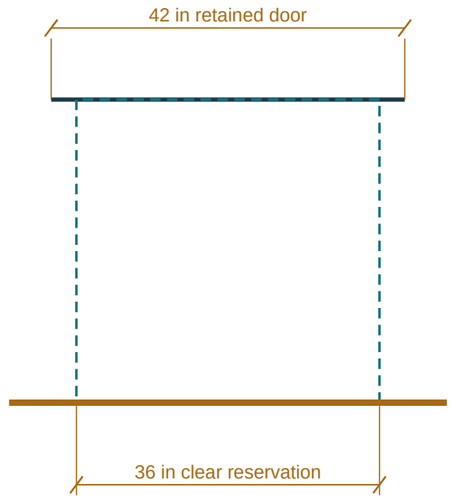


4 equal risers at7 in;3 treads at11 in.

Landing example zG0; floorz28 STUDY.

Rise=(28-zG)/N; run=(N-1)xgoing.

A3 / ACCESSIBLE SIDE DOOR / BRIDGE



Door x9..51; trial bridge widthx12..48.

Usable runB24 /36 /48 in trial cases.

For trial slope1:12: B >=12 xabs(deltaH).

B36 permits3 in vertical mismatch geometrically.

Bearing overlaps / restraint / load rating HELD.

A4 / MOVING INTERFACES

Stair hinge, lock, guards, handrail and stowed sweep require detailed engineering. Do not operate with deployed stairs or unsecured boarding equipment.

A5 / TRANSFER / RESCUE

Bridge requires rated supported bearing and retention at both interfaces. Independent stalled-train evacuation remains a separate design and training requirement.

A6 / GEOMETRY STATUS

Examples are not vehicle-code dimensions or construction authorization. Station height, actual suspension movement and emergency response determine the final system.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

PULSIPHER FARMS / CHRISTMAS RAILWAY

Hydraulic braking: local two-zone functional architecture

This is a control-function diagram, not a hydraulic circuit or valve schedule. The responsible brake supplier must release topology, sizes, settings and verified failure behavior.

H1 / DRIVER COMMAND / EMERGENCY PERMIT / LOCAL COACH CONTROLLER

D01 / Driver demand
Service command; integrity monitored

EM01 / Emergency permit
Loss / breakaway calls for application

BC01 / Coach brake controller
Command validation / zone balance / diagnostics. Emergency independence HELD.

PW01 / Protected power
Local control supply; outage behavior verified

TR01 / Traction permission
EM01 / fault / auxiliary release remove permission; reset HELD

AR01 / Keyed auxiliary release
Normally inactive and monitored. Auxiliary active inhibits traction.

To second release ports / topology HELD

status only

H2 / FRONT BOGIE ZONE

H01 / Pressure generation
Release-pressure regulation; reservoir, pump, filter and relief design HELD

PT01

SA01 / SA02 / Spring actuators
Spring force applies; working hydraulic pressure releases

PT03 auxiliary release pressure

EV01 / Emergency function
EM01 independent input; exact application topology HELD

XS: actuator position; WS: axle speed

TC: friction temperature; independent speed

H3 / REAR BOGIE ZONE

H02 / Pressure generation
Release-pressure regulation; reservoir, pump, filter and relief design HELD

PT02

SA03 / SA04 / Spring actuators
Spring force applies; working hydraulic pressure releases

PT04 auxiliary release pressure

EV02 / Emergency function
EM01 independent input; exact application topology HELD

XS: actuator position; WS: axle speed

TC: friction temperature; independent speed

EM01 independent application function / physical circuit HELD

H4 / CRITICAL DISTINCTION

Two hydraulic release ports on one spring caliper are NOT two independent stopping paths. An auxiliary release can defeat spring application unless correctly isolated and monitored. Exact safety function and failure coverage must be proven.

H5 / INTERCAR SERVICES

Proposed electric control / emergency trainline; no intercar hydraulic service line in this study. Separation must command brakes in all separated portions. Connector pinout, wire sizes, protocol and power reserve are held.

H6 / RELEASE TESTS

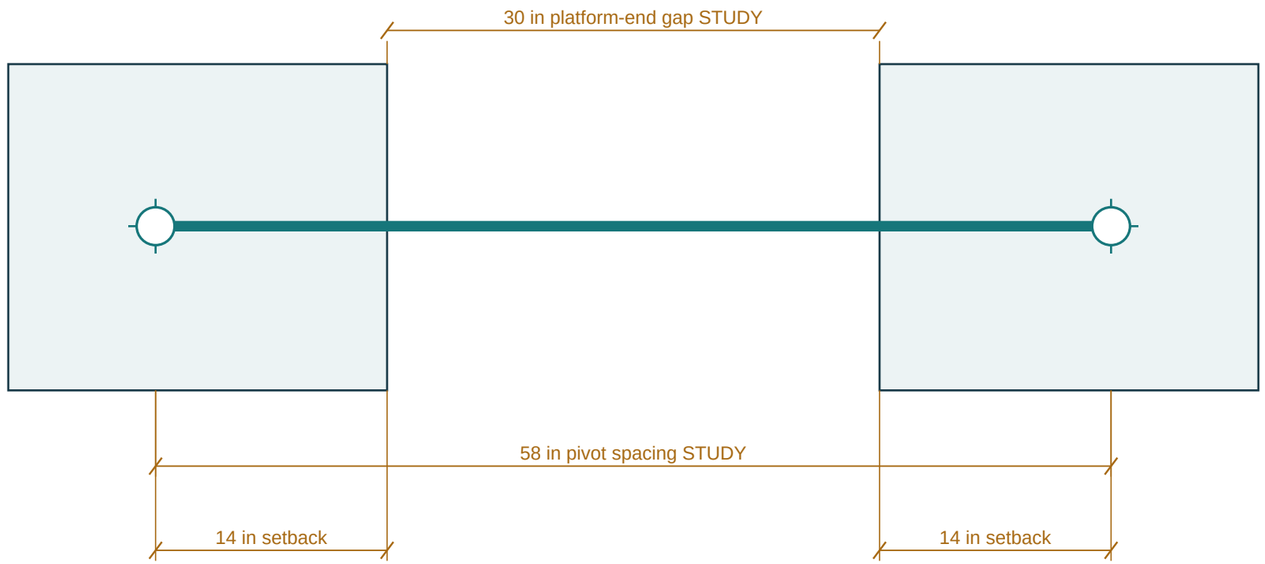
Verify broken wire, hose / pressure loss, stuck regulation, sensor disagreement, local supply failure and auxiliary-release misuse. Apply only engineer/OEM-approved tests at controlled pressure, energy and speed limits.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Connections: attachment load path and articulation study

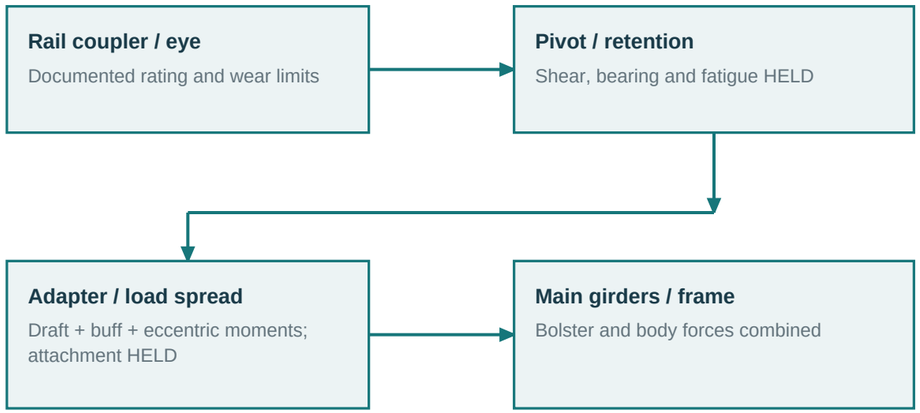
A railway drawbar / coupler package and engineered adapter are required. This drawing identifies force paths and checks; it does not size any pin, clevis, weld or retainer.

C1 / PLAN - RETAINED CONNECTION GEOMETRY



Guard and hose envelope: check combined yaw, pitch, roll, slack and vertical step.

C2 / LOAD PATH - ONE COACH END



Retention must remain effective under vibration, reversing and wear. Provide visual inspection access and a controlled replacement limit. An ordinary trailer tongue is not the design basis.

C3 / COORDINATED LONGITUDINAL LOAD SCREENS

Case / first joint where noted	Calculated screen	Status
Loco-only braking / first joint	16.579 kN compression	Retained3B case
Launch / first joint	12.552 kN tension	Retained3B case
Staggered brake response / max joint	6.275 kN compression	Rigid-body3C screen

Case inputs: loco-only mu0.15 and14,500 kg braked; launch a0.15 m/s2 +crr0.010. Staggered:0.8 m/s2 target; full latency / ramp data in controls calculations.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

C4 / WHAT IS NOT INCLUDED

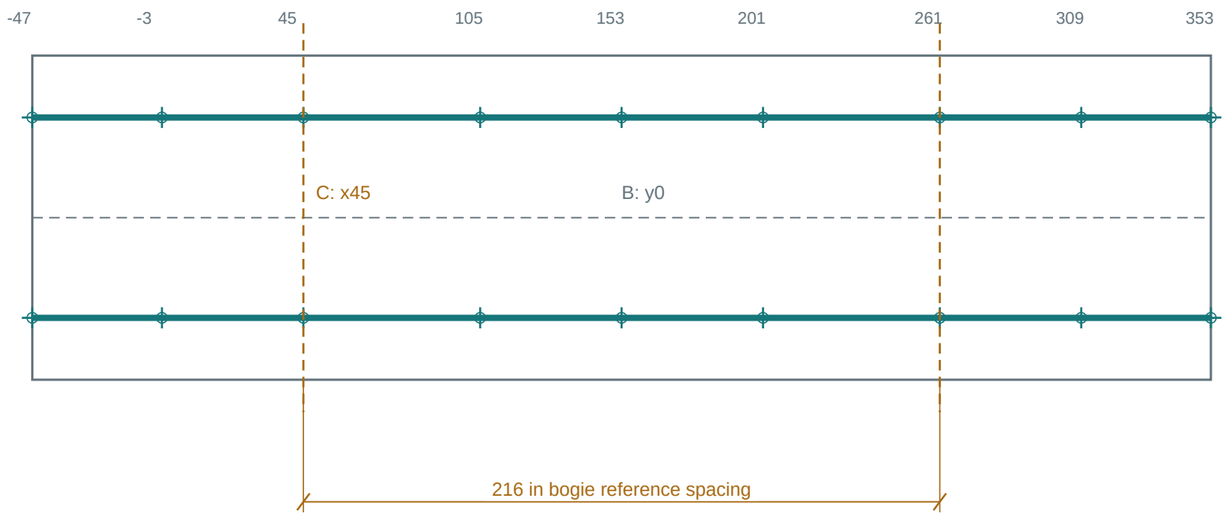
Slack take-up, buff impact, coupler eccentricity, derailment / rescue loads, fatigue spectrum and attachment stress concentration are not bounded by the table. These values cannot select a coupler or establish proof-test loads.

Require application acceptance and full 3D articulation envelope from the chosen supplier.

First frame: jig datum map and fabrication sequence

Shop datum scheme is proposed for review, not a released GD&T system. Jigs, stands, clamps, weld sequence and tolerances require the fabricator and responsible engineer.

J1 / CONTROL TARGETS - PLAN / NOT A STAND LAYOUT



18 targets =9 x stations by y=-34,+34. Survey target height from datumA.

A = floor-support plane zF0. Offset to physical targets is engineer-defined.

J2 / DATUM DEFINITION

A: frame floor-support plane; locate using surveyed jig targets with a defined offset. B: longitudinal centreline y0. C: front bogie crossline x45. Retain project x0 at front inner wall, 45 in forward of C.

Do not treat rail z0 as a shop support datum. Conversion requires accepted floor / bogie stack.

J3 / SUPPORT / CLAMP HOLD

The perimeter marks and 18 targets are measurement references, not approved support or jack points. Jig capacity, stand location, lateral restraint and clamp force are held.

Record support condition with every measurement. A free frame and a forcibly clamped frame cannot be compared as identical conditions.

J4 / CONTROLLED ASSEMBLY AND RECORDING SEQUENCE

T0 / Tool survey
Calibrate tools; verify jig / datum.

T1 / Dry fit
Record members, gaps, diagonals.

T2 / Tack
Qualified tack process; measure.

T3 / Weld groups
Engineer-approved numbered sequence.

T4 / Cool + release
Specified support state; survey / NDT.

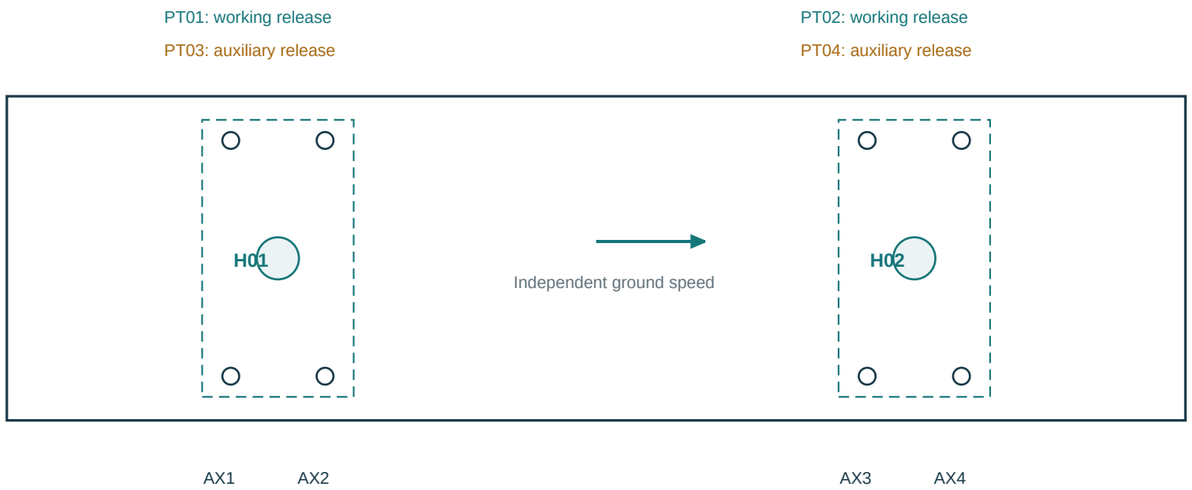
T5 / Test article
Before / after controlled ballast.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

First-article measurement map and acceptance records

Instrumentation defines what to measure; it does not authorize pressure, proof load or rolling tests. A responsible engineer / OEM must issue test limits, exclusion zones and stop rules.

V1 / ONE COACH - SENSOR / MEASUREMENT LOCATIONS



AX1..4: XS actuator position + WS axle speed + TC friction-temperature channels.

Use calibrated scales / load cells at approved supports; log four axle reactions.

Floor / frame targets: J01. Coupler axial load / motion: C01. Clock-sync all channels.

V2 / MEASURE BEFORE LOADING

Record configuration, serial numbers, tare, corner / axle loads, support conditions, CG assumptions, floor height, clearances, gauge / wheel geometry and controls baseline.

Instrumentation mounting must not weaken members, interfere with brakes or create passenger pinch points.

V3 / CONTROLLED TEST SEQUENCE

Check stationary function first. Then dead-load structural / floor tests and dimensional recovery under released procedures. Brake stand tests precede any rolling test.

Progress to unoccupied low-speed operation only after track, vehicle and test plan acceptance. Occupied demonstration is a later authorization.

V4 / RELEASE RECORDS / RESULTS MUST BE ENTERED, NOT PRE-CHECKED

Record IDs	Inspection / verification scope	Required record
I00-I03	Release, jig survey, dry fit, tack check	Drawing rev / tool ID / measured deviation / disposition
I04-I07	Weld groups, cooled geometry, NDT, coating	WPS / welder / inspection method / repair traceability
I08-I09	Bond coupons and body closure	Batch / preparation / cure / water and service-access checks
I10-I12	Ballast, brake / rolling, post-test examination	Calibrated loads / time histories / residuals / signed decision

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release